

# DERIVED PRISMATIC COHOMOLOGY

## 1. INTRODUCTION

So far, we have seen prismatic cohomology only on smooth  $p$ -adic formal schemes.

This assumption is essential! For example, in the proof of the Hodge-Tate comparison, even in characteristic  $p$  the étale localization step to a polynomial algebra was essential. You cannot ensure that the Frobenius on  $A/p$  is flat to get the untwisting via base change of prisms unless we do the localization to a polynomial algebra first.

The naive statements of the comparison theorems will not generalize to the non-smooth case, which we will amend by taking Kan extensions. For example, the de Rham comparison will be a comparison to derived de Rham cohomology, and so on.

## 2. KAN EXTENSION

The usual solution to this issue is to Kan extend everything that we have in the smooth case. The case we will deal with falls under the name of “non-abelian derived functors”.

Last semester, Daishi talked about the cotangent complex and how the formalism of non-abelian left derived functors can allow for a useful extension of  $\Omega^i$  past the smooth case. We will want to apply the same procedure to prismatic cohomology.

**DEFINITION 2.1 (Sifted colimits).** Let  $\mathcal{D}$  be a category. Then we call  $\mathcal{D}$  *sifted* if  $\mathcal{D}$ -colimits in  $\mathbf{Set}$  commute with finite products (for a proper  $\infty$ -categorical definition, you ask  $\mathcal{D} \rightarrow \mathcal{D} \times \mathcal{D}$  to be cofinal, i.e. precomposition with it preserves colimits).

A sifted colimit in a category  $\mathcal{C}$  is a colimit over a diagram  $\mathcal{D} \rightarrow \mathcal{C}$  where  $\mathcal{D}$  is sifted.

We write  $\mathrm{Fun}_{\Sigma}(-, -)$  for functors commuting with sifted colimits.

**EXAMPLE 2.2.** Any filtered category is sifted.

The opposite simplex category  $\Delta^{\mathrm{op}}$  is a sifted category which is not filtered. Thus, geometric realizations are sifted colimits.

To see this is sifted in the  $\infty$ -categorical sense, one shows  $\Delta^{\text{op}} \rightarrow \Delta^{\text{op}} \times \Delta^{\text{op}}$  is cofinal. For this, one proves the equivalent assertion that

$$\Delta^{\text{op}} \times_{\Delta^{\text{op}} \times \Delta^{\text{op}}} (\Delta^{\text{op}} \times \Delta^{\text{op}})_{[m] \times [n] /}$$

is contractible. This is the opposite category of the category  $\mathcal{C}$  of objects  $[p]$  with maps  $\{[p] \rightarrow [n], [p] \rightarrow [m]\}$ , so it suffices to show this is contractible. There's an adjunction (therefore a homotopy equivalence) to the category of monomorphisms  $[p] \rightarrow [m] \times [n]$  by mapping to  $\text{im } f \rightarrow [m] \times [n]$  where  $f : [p] \rightarrow [m] \times [n]$  is the original map in  $\mathcal{C}$  (and in the other way, the forgetful functor). It is possible to see the homotopy type of this final category is  $\Delta^n \times \Delta^m$ , so it is indeed contractible.

We are now ready to understand non-abelian derived functors.

**DEFINITION 2.3 (Non-abelian derived functors).** Let  $F : \text{Poly}_A \rightarrow \mathcal{C}$  be a functor landing in an  $\infty$ -category  $\mathcal{C}$  admitting all colimits. Taking left derived functors yields an equivalence

$$\text{Fun}(\text{Poly}_A, \mathcal{C}) \simeq \text{Fun}_{\Sigma}(\text{sAlg}_A, \mathcal{C}).$$

In other words, functors on polynomial algebras admit extensions to  $\text{sAlg}_A$  uniquely determined by the requirement that the extension commutes with sifted colimits.

The equivalence is given by sending  $F \mapsto LF$ , the left Kan extension of  $F : \text{Poly}_A \rightarrow \mathcal{C}$  along the inclusion  $\text{Poly}_A \subset \text{sAlg}_A$ .

Here, we use the  $\infty$ -category of simplicial commutative rings. In this setting, a simplicial commutative  $A$ -algebra is a presheaf  $\text{Poly}_A^{\text{op}} \rightarrow \mathcal{S}$  sending finite coproducts in  $\text{Poly}$  to products of spaces.

**REMARK 2.4.** For nice  $\mathcal{C}$  and  $\mathcal{D}$ , we clearly have

$$\text{Fun}(\mathcal{C}, \mathcal{D}) \simeq \text{Fun}_{\Sigma}(\mathcal{P}_{\Sigma}(\mathcal{C}), \mathcal{D})$$

where category  $\mathcal{P}_{\Sigma}(\mathcal{C})$  is given by *freely* adjoining sifted colimits. Formally, this is defined as the subcategory of presheaves (in spaces) on  $\mathcal{C}$  spanned by finite product preserving functors, but even from this perspective the claim is not difficult to prove.

By definition  $\mathcal{P}_{\Sigma}(\text{Poly}_A)$  yields  $\text{sAlg}_A$ , so the real content is that the  $\infty$ -categorical localization of the ordinary category of simplicial commutative rings yields this category.

To compute the total derived functor, one uses a projective resolution just like in the usual case with  $D(R)$ . Here, this takes the form

$$LF(B) \simeq |LF(P^\bullet)|$$

where  $P^\bullet \rightarrow B$  is some simplicial polynomial algebra resolution. Indeed, geometric realization  $|\cdot|$  is a sifted colimit, and by the determining requirement of  $LF$  the formula follows. Even with infinitely many generators,  $LF(A[S]) = \operatorname{colim}_{S' \subset S} LF(A[S'])$  over finite  $S'$  where we already know the value of the functor.

**EXAMPLE 2.5 (Abelian derived functors).** Let  $\mathcal{A}$  be a normal abelian category. The relevant objects are the compact projective objects  $P \in \mathcal{A}$ , for which  $\operatorname{Hom}(P, -)$  commutes with geometric realization of simplicial objects.

Assume  $\mathcal{A}$  has enough projectives. Then

$$\mathcal{P}_\Sigma(\mathcal{A}^{\operatorname{comp}, \operatorname{proj}}) \simeq D_{\geq 0}(\mathcal{A}).$$

The previous formalism taken with a functor  $\mathcal{A} \rightarrow \operatorname{Ab} \hookrightarrow D_{\geq 0}(\operatorname{Ab})$  is then *actually* computing the usual derived functor

**EXAMPLE 2.6.** Take  $\Omega_{B/A}^i$  as a functor  $\operatorname{Poly}_A \rightarrow D(A)$ . Then the cotangent complex  $\Lambda^i L_{B/A}$  is the nonabelian derived functor [3]. If  $i = 1$  we simply mean  $L_{B/A}$ , otherwise  $\Lambda^i$  actually means  $L\Lambda^i$ , the nonabelian derived functor of exterior powers.

An important fact we will use is that  $L_{B/A}$  agrees with  $\Omega_{B/A}$  when  $B$  is smooth; the similar statement for prismatic cohomology boils down to this fact. This follows from the fact that  $L_{B/A} = 0$  for étale  $A$ -algebras  $B$ , so using the transitivity triangle étale locally we are looking at the cotangent complex of polynomial algebras. More precisely, factor a smooth map as

$$A \rightarrow A[X_1, \dots, X_n] \rightarrow B$$

where the last map is étale. The transitivity triangle then yields an exact triangle

$$B \otimes L_{A[X_1, \dots, X_n]/A} \rightarrow L_{B/A} \rightarrow 0.$$

Thus, the first map is an isomorphism and we appeal to  $\Omega_{B/A}^1 \simeq B \otimes \Omega_{A[X_1, \dots, X_n]/A}^1$  by the similar exact sequence for Kähler differentials.

We are now ready for the definition of derived prismatic cohomology.

**DEFINITION 2.7 (Derived prismatic cohomology, [1, Construction 7.6]).** Let  $(A, I)$  be a bounded prism. The *derived prismatic cohomology* functor

$$R \mapsto L\Delta_{R/A}$$

on derived  $p$ -complete simplicial  $A/I$ -algebras  $R$  is defined as the non-abelian derived functor of  $\Delta_{R/A}$  on  $\text{Poly}_{A/I}^\wedge$ , with  $\mathcal{C}$  as the category of objects in  $D_{(p,I)\text{-comp}}(A)$  equipped with a  $\phi_A$ -semilinear endomorphism.

Specializing to derived  $p$ -complete classical rings, this admits a natural globalization  $\Delta_{\mathfrak{X}/A}$  to  $p$ -adic formal schemes.

**REMARK 2.8.** It is important that we use  $D_{(p,I)\text{-comp}}(A)$  rather than  $D(A)$ . Although on polynomial algebras the functor lands in this subcategory and the subcategory has all colimits, these colimits *do not* commute with the inclusion to  $D(A)$ .

In Bhatt's notes, he uses  $\text{Poly}_{A/I}$ , but this leads to some additional technical details later when it is no longer obvious  $L\Delta_{R/A} = \Delta_{R/A}$  on a  $p$ -completed polynomial ring. I've swapped this issue for hiding technical details in what happens for animated  $p$ -complete polynomial algebras.

Immediately after making this extension, we can obtain a derived Hodge-Tate comparison that will fix the defect in the previous example.

**LEMMA 2.9 (Derived Hodge-Tate, [1, Construction 7.6]).** The complex  $L\overline{\Delta}_{R/A}$  admits an exhaustive  $\mathbf{N}$ -indexed increasing filtration  $\text{Fil}_*^{\text{conj}}$  with

$$\text{gr}_i^{\text{conj}}(L\overline{\Delta}_{R/A}) \simeq (L\Lambda^i L_{R/\overline{A}}\{-i\}[-i])_p^\wedge$$

where the  $p$ -completion is derived.

*Proof.* Consider the case where  $R \in \text{Poly}_{A/I}^\wedge$ . We can put a truncation filtration on  $\overline{\Delta}_{R/A}$ , and via Hodge-Tate it has graded pieces<sup>1</sup>

$$\Omega_{R/(A/I)}^i\{-i\}[-i].$$

In particular, we can upgrade  $\overline{\Delta}_{R^\wedge/A}$  to an object in the filtered derived category

$$\text{DF}(A/I) := \text{Fun}(\mathbf{Z}^{\text{op}}, D_{\text{comp}}(A/I)).$$

We will still want to additionally enforce derived completeness to carry out the Kan extension, for the same reasons as the previous remark.

<sup>1</sup>As Kush pointed out this is why we call it the conjugate filtration: if we do the same thing in de Rham cohomology, it is opposite of the Hodge filtration.

Deriving this filtered enhancement, we obtain for general  $R$  a lift of  $L\bar{\Delta}_{R/A}$  in  $DF(A/I)$  (by this I mean you can forget the filtration and get the original object). Note that actually everything still gets an  $R$ -module structure; this is because it holds on polynomial rings, and we get a  $P^\bullet$ -module structure upon taking a resolution. This becomes an  $R$ -module structure.

Passing to underlying objects in  $D(A/I)$  or taking the associated graded  $gr_*$  in the filtered derived category preserves all colimits. Thus, the associated graded will have pieces

$$gr_i^{\text{conj}} \simeq L(\Omega_{-/(\bar{A})}^i \{-i\}[-i]),$$

which is almost what we want. One has to check that when we do the non-abelian derived functor procedure on  $\text{Poly}_{A/I}^\wedge$  instead of  $\text{Poly}_{A/I}$ , we get the derived  $p$ -completion of the cotangent complex. After this, the claim follows by noting that in this case (as well as the usual one without  $p$ -completions) derived functors of  $\Omega_{-/A}^i$  are  $L\Lambda^i L_{-/A}$ . One can check this by writing down polynomial resolutions to compute both, since on these we get projective modules after passing through  $L_{-/A}$  (we'll get polynomial algebras) and then  $L\Lambda^i$  will just be the usual exterior powers. So a polynomial resolution computes both of these functors as the same sifted colimit over  $\Delta^{\text{op}}$ .  $\square$

**PROPOSITION 2.10 (Consequences of the derived Hodge-Tate comparison, [1, Construction 7.6(1)–(5)]).** The following are also true for derived prismatic cohomology:

- On  $p$ -completely smooth algebras, the value of prismatic cohomology is unchanged.
- The functor  $R \mapsto L\Delta_{R/A}$  is a sheaf for the  $p$ -completely étale topology on the category of derived  $p$ -complete  $A/I$ -algebras.
- The functor  $R \mapsto L\Delta_{R/A}$  is also a sheaf for the quasisyntomic topology.
- The formation of  $L\Delta_{R/A}$  commutes with base change.
- The functor  $R \mapsto L\Delta_{R/A}$  commutes with all colimits; in particular, it is symmetric monoidal.

*Proof.* The idea for all of these is to use that  $L\Delta_{R/A}$  is derived  $(p, I)$ -complete, and  $-\otimes^L A/I$  is conservative (reflects isomorphisms). That is, we can test isomorphisms after passing to  $L\bar{\Delta}_{R/A}$  (this is essentially derived Nakayama).

We will show (1) in detail. To see that the value is unchanged on smooth algebras, it then suffices to check this for  $L\bar{\Delta}_{R/A}$  when  $R$  is  $p$ -completely smooth. By the universal

property of Kan extensions, we have a map from

$$L\overline{\Delta}_{R/A} \rightarrow \overline{\Delta}_{R/A}$$

compatibly with filtrations by using that this lifts to a morphism in  $DF(A/I)$ .

The associated graded functor  $gr_* : DF(A/I) \rightarrow D(A/I)$  is conservative. But we know the associated graded of both sides for a  $p$ -completely smooth algebra, and the map induced by the universal property will give the map

$$(L\Lambda^i L_{R/\bar{A}}\{-i\}[-i])^\wedge \rightarrow (\Omega_{R/\bar{A}}^i\{-i\}[-i])^\wedge$$

again induced by the universal property of Kan extensions. However, we know for the cotangent complex this is an isomorphism, and similarly  $L\Lambda^i : \text{Mod}_{A/I} \rightarrow D(A/I)$  does not change its values on flat modules (which will be the case here:  $\Omega^1$  is a projective module).

For quasisyntomic descent, by the same strategy the descent isomorphism can be reduced to testing for the cotangent complex which has flat descent.

The other claims are similar or easy to prove using the same general strategy to reduce to properties of the cotangent complex.  $\square$

I will now drop the  $L$ , as we know the values agree on smooth algebras.

In the previous counterexample to the comparison theorems, the value of prismatic cohomology will actually be unchanged. However, this Hodge-Tate comparison allows for the graded pieces to “pile up” in a single degree, allowing for infinite dimensional cohomology. It is in fact possible to explicitly describe the first component of this filtration.

**PROPOSITION 2.11 (Describing the first piece, [4, Proposition 3.2.1]).** There is a canonical isomorphism

$$\alpha_X : \text{Fil}_1^{\text{conj}} \simeq L_{X/A}\{-1\}[-1]^\wedge.$$

*Sketch.* This argument follows Anschütz and Le Bras.

First, we may assume  $X = \text{Spf}(R)$  is affine by descent. Using the transitivity triangle for  $A \rightarrow \bar{A} \rightarrow R$ , we obtain

$$R \otimes_{\bar{A}} L_{\bar{A}/A}\{-1\}[-1]^\wedge \longrightarrow L_{R/A}\{-1\}[-1]^\wedge \longrightarrow L_{R/\bar{A}}\{-1\}[-1]^\wedge$$

or just

$$gr_0^{\text{conj}} \longrightarrow L_{R/A}\{-1\}[-1]^\wedge \longrightarrow gr_1^{\text{conj}}.$$

It then already looks like the correct thing to consider (which hopefully removes some mystery from the statement).

To construct  $\alpha_X$ , it suffices to prove the case when  $\bar{A} \rightarrow R$  is  $p$ -completely smooth and Kan extend. We will therefore restrict ourselves to the smooth affine case in what follows.

Let us now define  $\alpha_X$ . Using smoothness, we essentially want to produce an isomorphism

$$\alpha_R : L_{R/A}^\wedge \rightarrow (\tau^{\leq 1} \bar{\Delta}_{R/A})\{1\}[1]$$

which can be done by producing a map to  $B/J\{1\}[1]$  for each prism in the prismatic site taking the colimit. Here, we're really using smoothness to ensure we can compute prismatic cohomology site-theoretically.

The first way of providing this map is very simple. Given an object  $(R \rightarrow B/J \leftarrow B)$  of the prismatic site  $(R/A)_\Delta$ , we have a natural map

$$L_{R/A}^\wedge \rightarrow L_{(B/J)/B}^\wedge$$

via the square

$$\begin{array}{ccc} R & \longrightarrow & B/J \\ \uparrow & & \uparrow \\ A & \longrightarrow & B. \end{array}$$

Note  $B/J$  cuts out an effective Cartier divisor or a regular codimension one embedding. In particular, it is lci. This yields

$$L_{(B/J)/B} \simeq J/J^2[1] \simeq B/J\{1\}[1].$$

Thus, we get the map! We can drop the completion since  $B$  defines a prism. Note that if  $R = A/I$ , we get the map from the base extension  $I/I^2 \rightarrow J/J^2$ .

There is also a way to write down this map in terms of Ext groups of the cotangent complex parameterizing deformation theory problems. For  $(B, J) \in (R/A)_\Delta$ , we have an extension

$$0 \longrightarrow J/J^2 \longrightarrow B/J^2 \longrightarrow B/J \longrightarrow 0.$$

By definition of the prismatic site, there is a map  $\iota : R \rightarrow B/J$ . Pulling this square-zero extension back along  $\iota$  gives a square-zero extension of  $R$  by  $J/J^2$  as an  $A$ -algebra:

$$0 \longrightarrow J/J^2 \longrightarrow E \longrightarrow R \longrightarrow 0.$$

The cotangent complex classifies such square-zero  $A$ -algebra extensions, so this extension determines an element of

$$\mathrm{Ext}_R^1(L_{R/A}, J/J^2) = \mathrm{Hom}_{D(R)}(L_{R/A}, J/J^2[1]) \simeq \mathrm{Hom}_{D(R)}(L_{R/A}, B/J\{1\}[1]).$$

That is, we again get a natural morphism

$$L_{R/A}^\wedge \rightarrow B/J\{1\}[1].$$

Taking a homotopy limit defines a map

$$\alpha_R : L_{R/A}^\wedge \rightarrow \tau^{\leq 0}(\overline{\Delta}_{R/A}\{1\}[1]) \simeq (\tau^{\leq 1}\overline{\Delta}_{R/A})\{1\}[1].$$

Really this is a fancy way of saying we took derived global sections; the truncation is because the cotangent complex lies in  $D^{\leq 0}$ . The cohomology sheaves of both sides are given by  $R\{1\}$  in degree  $-1$  and  $\Omega_{R/\bar{A}}^1$  in degree  $0$ . It suffices to show that  $\alpha_R$  induces the same maps as the Hodge-Tate comparison would on cohomology sheaves to get a functorial quasi-isomorphism in the smooth case.

We reduce to the case of  $A^1 = \mathrm{Spf}(\overline{A}\langle x \rangle)$ . This is enough, as it will also similarly prove the case of  $A^n$  (the only difference is you need to add more generators for  $\Omega^1$ ) and we can étale localize to this due to working in the smooth case. Base change can reduce to  $A = \mathbf{Z}_p$ , but this isn't necessary.

In this case, one explicitly has

$$L_{R/A}^\wedge \simeq (R \otimes_{\bar{A}} I/I^2)[1] \oplus \mathrm{Rd}x.$$

On the first summand, the map  $\alpha_R$  is easy to describe for a prism  $(B, J)$ : it is given by the base extension  $I/I^2 \rightarrow J/J^2$ . On  $\mathrm{Rd}x$ , the morphism factors as

$$R \xrightarrow{\iota} B/J \longrightarrow B/J\{1\}[1]$$

where the second morphism is the connecting morphism for  $0 \rightarrow B/J\{1\} \rightarrow B/J^2 \rightarrow B/J \rightarrow 0$ .

Now we pass to the limit with this description, and look at what happens for  $H^0$  (so we will study the  $\mathrm{Rd}x$  component). We obtain a diagram

$$\begin{array}{ccc} R & & \\ \downarrow \iota & \searrow & \\ \overline{\Delta}_{R/A} & \longrightarrow & \overline{\Delta}_{R/A}\{1\}[1] \end{array}$$

On  $H^0$ , by construction of the Bockstein the horizontal morphism induces

$$\beta_I : H^0(\overline{\Delta}_{R/A}) \rightarrow H^1(\overline{\Delta}_{R/A})\{1\}.$$

Thus, on  $H^0$  we map  $dx \mapsto \beta_I(\iota(x))$ . In particular,  $\alpha_R$  will induce the identity after making identifications  $H^0(L_{R/A}^\wedge) \simeq \Omega_{R/\bar{A}}^{1,\wedge}$  and  $H^0(\bar{\Delta}_{R/A}\{1\}[1]) \simeq \Omega_{R/\bar{A}}^{1,\wedge}$  (this is Hodge-Tate: the identification is  $f dx \mapsto f \beta_I(x)$ ).

Next, we check  $H^{-1}$ . This is easier, as  $\alpha_R$  is simply base extension and we see the map

$$H^{-1}(L_{R/A}^\wedge) \simeq R \otimes_{\bar{A}} I/I^2 \rightarrow H^{-1}(\bar{\Delta}_{R/A}\{1\}[1])$$

is the canonical one given by twisting the structure map  $R \rightarrow H^0(\bar{\Delta}_{R/A})$  (so in particular it also matches Hodge-Tate).

Thus, on cohomology sheaves we see that  $\alpha_R$  yields the same maps as the Hodge-Tate comparison. We can then conclude that  $\alpha_R$  induces the canonical Hodge-Tate comparison isomorphisms on cohomology sheaves for general smooth algebras, which concludes the proof.  $\square$

**THEOREM 2.12 (Discrete examples, [1, Lemma 7.7]).** Assume that  $(A, I)$  is a bounded prism, and assume that  $R$  is a derived  $p$ -complete simplicial  $A/I$ -algebra such that  $\bar{\Delta}_{R/A}$  is concentrated in degree zero.

The following hold:

- (1) The  $\varphi_A$ -linear Frobenius  $\varphi_R$  on  $\Delta_{R/A}$  naturally lifts to a  $\delta$ - $A$  structure on the ring  $\Delta_{R/A}$ .
- (2) The pair  $(\Delta_{R/A}, I\Delta_{R/A})$  gives a prism over  $(A, I)$  equipped with a map

$$R \rightarrow \bar{\Delta}_{R/A} = \Delta_{R/A}/I\Delta_{R/A}.$$

- (3) The category “ $(R/A)_\Delta$ ” of prisms  $(B, J)$  over  $(A, I)$  equipped with a map

$$R \rightarrow B/J$$

has an initial object. Moreover, the initial object is the image of an idempotent endomorphism of  $(\Delta_{R/A}, I\Delta_{R/A})$ .

*Proof.* First, note that  $\Delta_{R/A}$  is also concentrated in degree zero by applying derived Nakayama to the map  $H^0(\Delta_{R/A}) \rightarrow \Delta_{R/A}$ .

For item (1), it will be easiest to interpret a  $\delta$  structure on a simplicial ring  $\mathcal{R}$  as map  $w : \mathcal{R} \rightarrow W_2(\mathcal{R})$  so  $\varepsilon \circ w = \text{id}$  ( $\varepsilon$  drops the second coordinate of  $W_2$ , so  $w(x) = (x, \delta(x))$ ).

To obtain a simplicial version of the Čech-Alexander complex, first consider a polynomial resolution  $P^\bullet$  of the simplicial  $A/I$ -algebra  $R$ . This can be done functorially. We then have

$$\Delta_{R/A} \simeq |\Delta_{P^\bullet/A}| = \text{colim}_{\Delta^{\text{op}}} \Delta_{P^\bullet/A}.$$

where the polynomial  $A/I$ -algebras in  $P^\bullet$ , if chosen functorially, usually have infinitely many generators. Prismatic cohomology on these is a colimit of cohomologies of finitely generated polynomial algebras.

We have a Čech-Alexander complex computing  $\Delta_{P^\bullet/A}$ , which again can be done functorially. These are *cosimplicial* algebras, and to obtain the actual complex one takes totalization: this is a limit over  $\Delta$ .

It follows that

$$\Delta_{R/A} \simeq \operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} F_A(R)$$

where  $F_A(R)$  is a simplicial object that assembles the Čech-Alexander complexes for  $P^\bullet$ . That is,  $F_A(R)$  is a *simplicial cosimplicial* derived  $(p, I)$ -complete  $\delta$ - $A$ -algebra. The cosimplicial comes from the Čech-Alexander complexes for individual  $P^i$  which are cosimplicial  $\delta$ - $A$ -algebras, but these  $P^i$  come from a simplicial polynomial algebra  $P^\bullet$  and so we assemble them into a simplicial cosimplicial object.

The point of all of this is that  $F_A(R)$  has a  $\delta$ - $A$ -algebra structure classified by a map

$$F_A(R) \rightarrow W_2(F_A(R)).$$

On the right, this means you apply  $W_2$  termwise as everything is termwise a  $\delta$ - $A$ -algebra.

This morphism lies over  $A \rightarrow W_2(A)$  and splits  $W_2(F_A(R)) \rightarrow F_A(R)$  (this is the  $\varepsilon$  part of the characterization). Now just applying colimits and limits to everything, we get a map

$$\Delta_{R/A} \rightarrow \operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} W_2(F_A(R))$$

lying over  $A \rightarrow W_2(A)$ , and splitting

$$\operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} W_2(F_A(R)) \rightarrow \Delta_{R/A}.$$

**Claim.** We have  $\operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} W_2(F_A(R)) \simeq W_2(\Delta_{R/A})$ .

Put

$$X_\bullet := \lim_{\Delta} F_A(R), \quad |X_\bullet| \simeq \Delta_{R/A}.$$

To prove the claim, use the functorial pullback square

$$\begin{array}{ccc} W_2(B) & \xrightarrow{F} & B \\ \downarrow & & \downarrow \\ B & \xrightarrow{\varphi} & B \otimes_{\mathbb{Z}}^L \mathbb{F}_p \end{array}$$

which also holds in simplicial rings. After applying totalization in the cosimplicial direction, the resulting square remains cartesian because limits commute with limits; moreover,  $- \otimes_{\mathbb{Z}}^L \mathbb{F}_p$  commutes with these totalizations because  $\mathbb{F}_p$  is perfect as a

complex of  $\mathbf{Z}$ -modules. Thus  $\lim_{\Delta} W_2(F_A(R))$  identifies with  $W_2(X_{\bullet})$  degree-wise in the remaining simplicial direction.

Let us spell out why animated  $W_2$  preserves the geometric realization in this last simplicial direction. First check the corresponding statement on ordinary rings. Let  $A_{\alpha}$  be a sifted diagram of ordinary rings, and consider the comparison map

$$\operatorname{colim}_{\alpha} W_2(A_{\alpha}) \longrightarrow W_2(\operatorname{colim}_{\alpha} A_{\alpha}).$$

Sifted colimits of ordinary commutative rings are created by the underlying set functor, and that functor reflects isomorphisms. After applying it, the comparison becomes

$$\operatorname{colim}_{\alpha} (A_{\alpha} \times A_{\alpha}) \longrightarrow (\operatorname{colim}_{\alpha} A_{\alpha}) \times (\operatorname{colim}_{\alpha} A_{\alpha}),$$

because  $W_2(A)$  has underlying set  $A^2$ . This map is a bijection exactly because sifted colimits commute with finite products in sets. Thus  $W_2$  preserves sifted colimits on ordinary rings. Its animation is the sifted-colimit-preserving left Kan extension from polynomial rings, so it preserves geometric realizations. Consequently

$$\operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} W_2(F_A(R)) \simeq |W_2(X_{\bullet})| \simeq W_2(|X_{\bullet}|) \simeq W_2(\Delta_{R/A}),$$

where the last term is the usual length-two Witt vector ring since  $\Delta_{R/A}$  is discrete. This proves the claim.

Under this identification, after realization the displayed square is the defining pull-back square for  $W_2(\Delta_{R/A})$ :

$$\begin{array}{ccc} \operatorname{colim}_{\Delta^{\text{op}}} \lim_{\Delta} W_2(F_A(R)) & \xrightarrow{F} & \Delta_{R/A} \\ \downarrow & & \downarrow \\ \Delta_{R/A} & \xrightarrow{\varphi} & \Delta_{R/A} \otimes_{\mathbf{Z}}^{\mathbf{L}} \mathbf{F}_p. \end{array}$$

Here, the bottom right corner is identified using that  $- \otimes_{\mathbf{Z}}^{\mathbf{L}} \mathbf{F}_p$  commutes with geometric realizations. Revisiting the maps we wrote earlier with this identification, by discreteness we have precisely produced a  $\delta$ - $A$ -algebra structure on  $\Delta_{R/A}$ , proving (1).

Part (2) is then immediate from derived  $(p, I)$ -completeness and discreteness of  $\overline{\Delta}_{R/A}$ . To deduce the claim, we need  $\Delta_{R/A}$  to be a discrete derived  $(p, I)$ -complete  $\delta$ - $A$ -algebra without  $I$ -torsion to apply Lemma 3.5 of Bhatt-Scholze's *Prisms and prismatic cohomology* [1, Lemma 3.5] (rigidity of prisms). Observe  $\Delta_{R/A} \otimes_{\mathbf{A}}^{\mathbf{L}} A/I^n$  must also be discrete after reducing to  $I = (d)$  and applying the distinguished triangle

$$\Delta_{R/A} \otimes_{\mathbf{A}}^{\mathbf{L}} A/I^n \longrightarrow \Delta_{R/A} \otimes_{\mathbf{A}}^{\mathbf{L}} A/I^{n+1} \longrightarrow \overline{\Delta}_{R/A}.$$

Since the Mittag-Leffler condition is satisfied the  $R \lim$  will be discrete. We don't get any  $I$ -torsion since reducing to  $I = (d)$ , we see  $H^{-1}(\bar{\Delta}_{R/A})$  is  $\mathrm{Tor}_1^A(A/d, \Delta_{R/A})$  which is isomorphic to the  $d$ -torsion.

For part (3) we need to do slightly more work. We will first construct a functorial map

$$(\Delta_{R/A}, I\Delta_{R/A}) \rightarrow (B, J)$$

for any  $(B, J) \in (R/A)_\Delta$  (we use the same meaning as in the statement of (3); recall  $R$  need not be a discrete ring here).

Fix a resolution  $P^\bullet \rightarrow R$  of  $p$ -completely ind-smooth  $A/I$ -algebras (we can even take these to be ind-polynomial). Now we have

$$\Delta_{R/A} = \mathrm{colim}_{\Delta^{\mathrm{op}}} \lim_{C \in (P^\bullet/A)_\Delta} C$$

in the category of derived  $(p, I)$ -complete complexes. This is because  $\lim_{C \in (P^i/A)_\Delta} C$  computes  $\Delta_{P^i/A}$  in the smooth affine case, as we may use the indiscrete topology (really Čech-Alexander theory tells us we get the same answer).

Now  $(B, J)$  (like any other prism we are considering in  $(R/A)_\Delta$ ) yields compatible objects in  $(P^\bullet/A)_\Delta$ , which by abuse of notation we all denote by  $B$ . This induces a map

$$\Delta_{R/A} \simeq \mathrm{colim}_{\Delta^{\mathrm{op}}} \lim_{C \in (P^\bullet/A)_\Delta} C \rightarrow \mathrm{colim}_{\Delta^{\mathrm{op}}} B = B.$$

Repeating the argument for (1) shows we get a map of  $\delta$ -rings. Thus, we have shown that  $(\Delta_{R/A}, I\Delta_{R/A})$  is weakly initial.

To see we get an initial object via an idempotent endomorphism, we appeal to the following categorical lemma.

**LEMMA 2.13** ([1, Lemma 7.8]). Let  $\mathcal{C}$  be an idempotent complete category and  $X \in \mathcal{C}$  an object. Let  $\mathcal{C}_{X\setminus}$  be the category of objects  $Y$  with a morphism  $X \rightarrow Y$ .

Assume  $\mathrm{id}_{\mathcal{C}}$  factors over

$$\mathcal{C}_{X\setminus} \rightarrow \mathcal{C}$$

via  $F : \mathcal{C} \rightarrow \mathcal{C}_{X\setminus}$ . Then  $F(X)$  yields an idempotent endomorphism of  $X$ , and the corresponding retract (image) is an initial object of  $\mathcal{C}$ .

Apply this to  $\mathcal{C} = (R/A)_\Delta$  and  $X = (\Delta_{R/A}, I\Delta_{R/A})$ . The functorial construction above assigns to each object  $(B, J)$  a morphism  $X \rightarrow (B, J)$ , hence gives the required factorization of  $\mathrm{id}_{\mathcal{C}}$  through  $\mathcal{C}_{X\setminus}$ .  $\square$

Finally, we will end with an example of computing prismatic cohomology.

Let  $(A, I)$  be a bounded prism, and set

$$R := A/(I, f_1, \dots, f_r)$$

where  $f_i$  are Koszul-regular on  $A/I$ . Assume this has bounded  $p$ -torsion.

**Claim.**  $\Delta_{R/A}$  is concentrated in degree 0 and  $I$ -torsionfree.

This follows from the Hodge-Tate comparison. We know we have a conjugate filtration of  $\overline{\Delta}_{R/A}$  given by  $\Lambda^i L_{R/(A/I)} \{-i\}[-i]^\wedge$ . As  $R$  is given by a quotient of  $A/I$  by a Koszul-regular sequence  $f_i$  (generating  $J = (f_i)$ ), we have

$$L_{R/(A/I)} \simeq (J/J^2)[1].$$

Looking at the derived exterior powers, we'll get  $\Gamma_{A/I}^i(J/J^2)[i] = ((J/J^2)^{\otimes i})^{\Sigma_i}$ . Thus all of these components of the conjugate filtration will be concentrated in degree zero, and the claim follows.

Now we are in the situation of the previous theorem, and so  $(\Delta_{R/A}, I\Delta_{R/A})$  is a weakly initial object in  $(R/A)_\Delta$ .

**PROPOSITION 2.14** ([1, Example 7.9]). Let  $R = A/(I, f_1, \dots, f_r)$  where  $f_i$  are Koszul-regular on  $A/I$  as before. The prism  $(\Delta_{R/A}, I\Delta_{R/A})$  is initial in  $(R/A)_\Delta$ . Equivalently,  $\Delta_{R/A}$  is the prismatic envelope of  $(A, J)$  for  $J = (I, f_1, \dots, f_r)$ ; Zariski locally, if  $I = (d)$ , this is  $A \left\{ \frac{f_i}{d} \right\}_{(p,d)}^\wedge$ .

**REMARK 2.15.** When the prism is perfect, this example is particularly important: this shows that the prismatic cohomology of a quasiregular semiperfectoid ring is  $\mathbf{A}_{\text{inf}}(R) \left\{ \frac{f_i}{d} \right\}^\wedge$ .

*Proof.* This is a local assertion on  $\text{Spf } A$ , so by ind-Zariski localization we may assume that  $I = (d)$ . Define

$$B := A \left\{ \frac{f_i}{d} \right\}_{(p,d)}^\wedge$$

We have a natural map  $B \rightarrow \Delta_{R/A}$ ; specifying a map from  $B$  into a  $d$ -torsionfree  $\delta$ - $A$ -algebra  $\Delta_{R/A}$  amounts to specifying a map  $A \rightarrow \Delta_{R/A}$  (the structure map) so that it carries the  $f_i$  into  $d\Delta_{R/A}$ . In  $\overline{\Delta}_{R/A}$ , by the previous calculation the  $f_i$  vanish. Thus we indeed have a map.

If we can show that the natural map

$$B \rightarrow \Delta_{R/A}$$

is an isomorphism, we learn that  $B$  is discrete so  $(B, dB) \in (R/A)_\Delta$ . We also learn that this prism  $(B, dB)$  must be initial. Given a prism  $(A', dA')$  in  $(R/A)_\Delta$  (so it has a map  $R \rightarrow A'/d$ ) to specify a map  $(B, dB) \rightarrow (A', dA')$  amounts to a map  $A \rightarrow A'$  sending the  $f_i$  into  $(d)$ . The existence of  $R = A/(d, f_1, \dots, f_r) \rightarrow A'/d$  forces the condition on the  $f_i$  to hold, so there is a unique map  $A \rightarrow A'$  induced by this. Thus, such an isomorphism would tell us that the initial object of  $(R/A)_\Delta$  is  $(B, dB) \simeq (\Delta_{R/A}, d\Delta_{R/A})$ , which is everything we wanted.

Now we turn to showing  $B \rightarrow \Delta_{R/A}$  is an isomorphism. It suffices to check in the universal case by base change. Namely, you can put  $A = \mathbf{Z}_p\{d, f_1, \dots, f_r\}[\delta(d)^{-1}]^\wedge$ . Here  $A/d$  and  $R$  are  $p$ -torsionfree.

In this case  $B = A \left\{ \frac{f_i}{d} \right\}_{(p,d)}^\wedge$  identifies as the prismatic envelope of  $(A, J = (d, f_1, \dots, f_r))$ , after checking the  $(p, d)$ -complete regularity condition. It follows  $B$  is discrete, and by the same reasoning it is initial.

By part (3) of the previous theorem, the map  $B \rightarrow \Delta_{R/A}$  admits a retraction (i.e. we get  $B \rightarrow \Delta_{R/A} \rightarrow B$  with composite equal to the identity). It follows  $B/dB \rightarrow \overline{\Delta}_{R/A}$  also admits a retraction. The cokernel of this map will be  $p$ -torsionfree as  $\overline{\Delta}_{R/A}$  is (since  $R$  is). We may therefore just check that

$$(B/dB)[1/p] \twoheadrightarrow \overline{\Delta}_{R/A}[1/p],$$

as this will imply the cokernel is trivial and then the claim follows.

By Hodge-Tate, as it is discrete we know  $\overline{\Delta}_{R/A}[1/p]$  is generated as a (Banach)  $R[1/p]$ -algebra by  $\text{gr}_1 = J/(J^2 + I)\{-1\}$ . It is therefore sufficient to show that

$$B = A \left\{ \frac{J}{I} \right\} \rightarrow \Delta_{R/A}$$

maps  $\frac{J}{I}$  surjectively onto  $\text{gr}_1$ . This is not too hard to check, since you basically need to see that  $\frac{f_i}{d}$  don't get sent to zero.  $\square$

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